AI to Support Pharmacogenetics in Understudied Populations

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Introduction

Background:

- Regulatory agencies in the United States oversee the safety and efficacy of therapeutics and diagnostics for a diverse population
- This responsibility becomes particularly challenging in the context of genetic testing and genetically informed therapeutics, where product performance can vary due to genetic differences across populations
- **Pharmacogenomics (PGx)** studies how genetic variations affect drug response, revealing population differences in allele distribution (Fig. 1). PGx offers actionable insights for personalized treatment and is increasingly clinically implemented [1, 2]

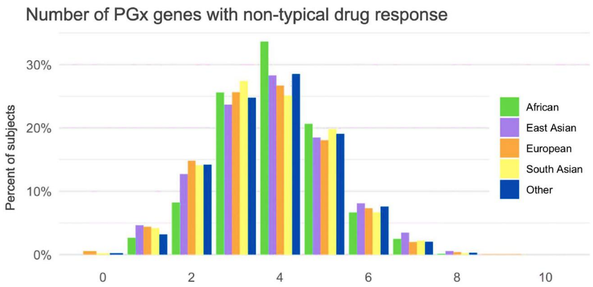

Fig. 1. The distribution of nontypical response alleles in PGx genes across populations [1]

Motivation:

- Despite growing interest, most genetic and PGx studies have focused on populations of European genetic ancestry, leaving significant gaps in understanding in non-European ancestry groups (Fig. 2) [3, 4]

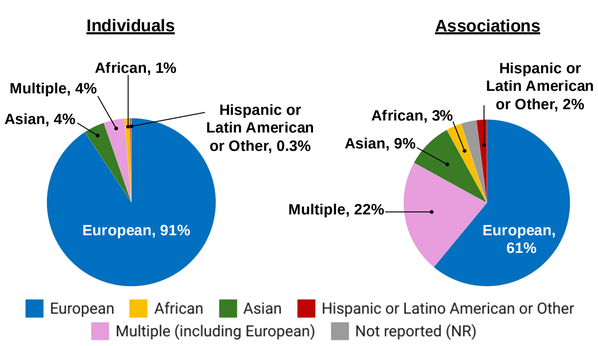

Fig. 2. Distribution of ancestry category labels as percentages based on Individuals (N = 6,373,952,077) and Associations (N = 624,819) up to 1st July 2024. Data adapted from [4]

- Biobanks such as All of Us [5], Million Veteran Program (MVP) [6], and UK Biobank [7], which have accumulated substantial health, lifestyle, exposure, and genetic information from non-European ancestry populations, provide an opportunity to address these gaps (Table 1)

	Whole Genome Sequencing Data				Genotype Array Data			
	UK Biobank	All of Us*	MVP**	N	UK Biobank	All of Us*	MVP	N
African	9,229	50,969	21,099	81,297	6,636	63,645	123,120	193,401
American Admixed/ Latino	NA	47,371	5,596	52,967	980	59,853	52,183	113,016
Asian	11,919	7,647	789	20,355	11,585	9,892	8,329	29,806
European	458,855	129,525	63,262	651,642	420,531	166,739	464,961	1,052,231
Middle Eastern	NA	1,389	NA	1,389	1,599	1,846	NA	3,445
Other	10,637	7,048	7,754	25,439	NA	9,094	9,989	19,083
Total	490,640	238,516	98,500	827,656	441,331	304,121	658,582	1,404,034

* self-assessed ancestry
** only subset of the whole available data was analyzed for ancestry, whole data is ~140 000 sequences, as of Feb 2023

Table 1. Broad race/ethnic group breakdown of whole genome sequencing and genotyped array data in three biobanks. Counts obtained from [8, 9, 10]

Proposal:

- **We propose to help close the knowledge gap about PGx in understudied populations by leveraging genetic and phenotypic data in biobanks and advances in AI methods**

Expected Outcomes

- Improved knowledge of PGx variations, especially in non-European groups, will help regulators make more informed decisions on genetics-based therapies
- Developed computational tools and methods to improve assessment of new products influenced by genetic variations
- Facilitate more personalized treatment strategies and promote equitable access to precision medicine
- Broader application to diagnostics, protein biologics, and vaccine development

Approach

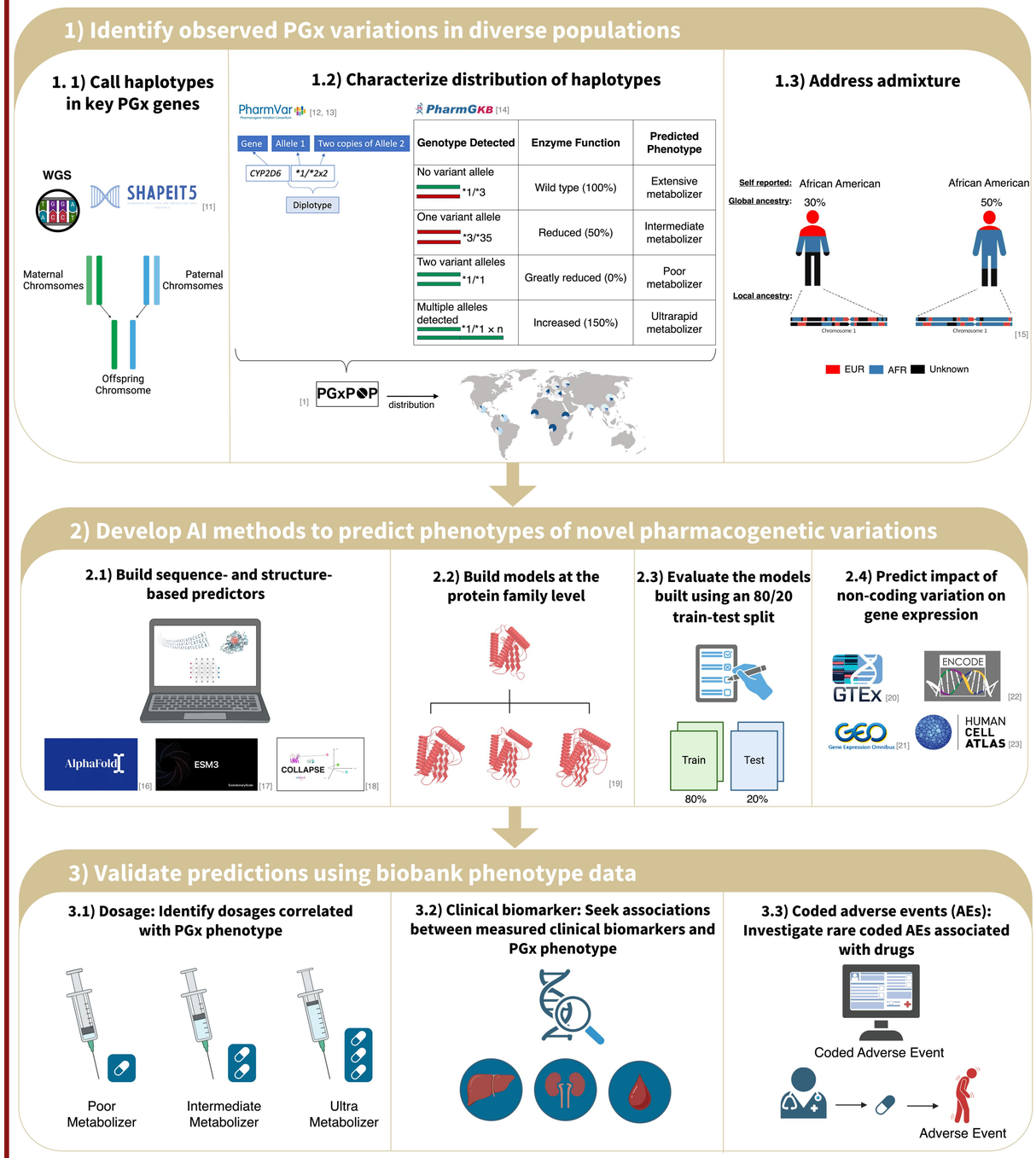

Fig. 3. Approach overview. Components adapted from external resources are labeled accordingly within the figure. All other elements were created using BioRender.com

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